

RISK MANAGEMENT FILE

CSI-100 Cell Sorting Instrument (IVD)

Document number	RMF-CSI-001 (includes HA-CSI-001 Rev D, RMR-CSI-001 Rev B)
Revision	D (living file; supersedes Rev C)
Document type	Risk management file (ISO 14971:2019)
Status	Released
Effective date	December 2028 (first annual post-market update)
Author	Atif Lodhi, Systems Engineering
Product	CSI-100 Cell Sorting Instrument, all configurations

Purpose

End-to-end risk management for the CSI-100 per ISO 14971:2019: plan, hazard identification, risk estimation and evaluation, risk controls and their verification, residual risk, benefit-risk analysis, risk management report, and production and post-production activities. Maintained for the product life.

Document Control

Rev	Date	Phase	Description of change	Author
A	Dec 2024	Concept	Risk management plan; preliminary hazard analysis HA-01 to HA-12.	A. Lodhi
B	Jan 2026	Design	Added HA-13 to HA-15; HA-03 merged into HA-04; DFMEA; software risk analysis; risk controls allocated to SRS-CSI-001 Rev C.	A. Lodhi
C	May 2027	Pre-market release	Added HA-16 and HA-17; PFMEA; verification of risk controls; V&V anomalies evaluated; overall residual risk; benefit-risk analysis; risk management report RMR-CSI-001 Rev A.	A. Lodhi
D	Dec 2028	Post-market (first annual review)	Hazard analysis restructured to 36 hazardous situations with separate P1 and P2 estimates and documented basis (internal audit finding IA-2028-07). Post-market data for 2028 evaluated; HS-08.2 P1 revised; HS-17.2 added; controls from CAPA-2028-014 added (FW-311, RA-107, SYS-327). Use-related risk, software, security (CVSS) and FMEA sections expanded. RMR-CSI-001 Rev B.	A. Lodhi

Summary of changes in Rev D

Change	Reason	Sections
Each hazard decomposed into one or more hazardous situations (HS-xx.y) with a sequence of events, P1, P2 and written basis for each estimate	IA-2028-07: Rev C estimated probability per hazard, which did not show how each control reduced P1 or P2	6, 7
HS-08.2 P1 revised from 1×10^{-2} to 1×10^{-1}	Observed thermal drift on 7 of 64 instruments in 2028	6, 8.2, 21
HS-17.2 added (instrument unavailable after repeated QC failure)	Not analyzed before; 41 QC-blocked events at 7 sites	6, 10, 21
New controls FW-311, RA-107, SYS-327 verified	CAPA-2028-014 corrective actions CA1, CA7, PA2	6, 7, 12
Harms catalogue with severity rationale; aggregated risk per harm	Consistency of severity across hazards; ISO/TR 24971 guidance on overall residual risk	5.2, 17
URRA with PCA analysis; CVSS scores before and after controls	FDA human factors and premarket cybersecurity guidance	13, 14

Approvals

Function	Name	Role	Signature / Date
Quality Engineering	M. Serrano	Risk management owner	
Systems Engineering	A. Lodhi	Author; technical lead	
Clinical / Medical Affairs	Dr. K. Brandt	Clinical risk and benefit assessment	
R&D Engineering	J. Whitfield	Hardware risk controls	
Software Engineering	P. Nair	Software and security risk	
Human Factors	L. Osei	Use-related risk	
Manufacturing Engineering	D. Kowalski	Process risk (PFMEA)	
Top management	VP Quality and Regulatory	Risk acceptability policy; overall residual risk	

Contents

1. Introduction, Scope and File Structure
2. Risk Management Plan Summary
3. Intended Use and Safety Characteristics
4. Risk Analysis Methods and Criteria
5. Hazard Identification and Harms
6. Hazard Analysis and Risk Control (landscape tables)
7. Verification of Risk Controls
8. Worked Examples of Risk Estimation
9. Risk Profile Before and After Control
10. Risks Arising from Risk Controls and Completeness
11. Software Risk Management
12. FMEA Summaries (landscape)
13. Use-Related Risk Analysis
14. Cybersecurity Risk
15. Risks from Anomalies Found in V&V
16. Aggregated Risk per Harm
17. Overall Residual Risk Evaluation
18. Benefit-Risk Analysis
19. Disclosure of Residual Risks
20. Risk Management Report
21. Production and Post-Production Activities
22. Reference Documents

1. Introduction, Scope and File Structure

This Risk Management File (RMF) records risk management for the CSI-100 Cell Sorting Instrument according to ISO 14971:2019 and ISO/TR 24971:2020. It covers the complete system: benchtop sorter, fluidics supply module, workstation, instrument software and firmware, plate and slide deposition option, aerosol management option, accessories and consumables supplied by the manufacturer. Reagents and assays are covered by their own risk management files; their interfaces with the instrument (QC beads, locked assay templates) are covered here.

The file is maintained for the life of the product. Rev D is the first annual update after commercial release in June 2027 and includes the post-market review of calendar year 2028 (installed base of 64 instruments, about 58 instrument-years of use).

ISO 14971 clause	Content	Location
4.1 to 4.5 General requirements	Process, management responsibility, competence, plan, file	Section 2; RMP-CSI-001 Rev C
5.2, 5.3 Intended use; safety characteristics	Intended use, users, environment, misuse, characteristics	Section 3
5.4 Hazards and hazardous situations	Hazard list, sequences of events, harms	Sections 5 and 6
5.5 Risk estimation	P1, P2 and severity with basis	Sections 4, 6 and 8
6 Risk evaluation	Acceptability of each risk	Section 6 Table 6-1
7.1 to 7.4 Risk control	Options analysis, implementation, residual risk, benefit-risk	Sections 6, 7, 18
7.5, 7.6 New risks; completeness	Risks arising from controls; completeness check	Section 10
8 Overall residual risk	Evaluation method and conclusion	Sections 16, 17
9 Risk management review	Risk management report RMR-CSI-001 Rev B	Section 20
10 Production and post-production	Information collection, review and actions	Section 21

1.1 Terms used in this file

Term	Meaning in this file
Hazard (HA-xx)	Potential source of harm, e.g. Class 3B laser radiation.
Hazardous situation (HS-xx.y)	Circumstance in which a person is exposed to a hazard, reached by a described sequence of events.
Harm (H-xx)	Injury or damage to health of a patient, operator or service engineer. Each harm has one severity (Section 5.2).
P1	Probability that the hazardous situation occurs, per instrument-year of use.
P2	Probability that the hazardous situation leads to the harm.
P	Probability of harm, $P = P1 \times P2$, assigned to a probability level P1 to P5 (Section 4.2).
Initial risk / residual risk	Risk before any risk control of this analysis / after all listed controls are in place and verified.

2. Risk Management Plan Summary

The full plan is RMP-CSI-001 Rev C. The elements that govern this file are summarized here.

2.1 Risk acceptability policy

The policy is set by top management in QP-RM-001 and applied to the CSI-100 in RMP-CSI-001. Risks are reduced as far as possible without adversely affecting the benefit-risk ratio. A risk in the unacceptable region is not released. A risk in the review region is accepted only when further reduction is not practicable and a documented benefit-risk analysis shows that the benefit outweighs the risk. The criteria take into account the state of the art for flow cytometers and droplet cell sorters (published adverse event data, recalls and the labeling of similar devices), the applicable product standards and the expectations of laboratory users.

2.2 Team and competence

Role	Competence relevant to this file	Evidence
Risk management owner (Quality)	ISO 14971 practitioner; 12 years medical device QA	Training record TR-0412
Systems engineer (author)	Flow cytometry instrument design; system hazard analysis	CV on file; TRN-RM-002
Clinical affairs	MD, hematopathology; clinical flow cytometry interpretation	CV on file
Laser safety officer	IEC 60825-1; certified laser safety officer	Certificate LSO-2025-117
Electrical safety	IEC 61010-1 test and design experience	CV on file
Biosafety	Biosafety professional; cell sorter aerosol containment testing	Certificate RBP-8841
Software and security	IEC 62304, IEC/TR 80002-1, threat modeling (IEC 81001-5-1)	Training records
Human factors	IEC 62366-1; summative study design	CV on file
Manufacturing engineering	PFMEA; process validation	Training records
Independent reviewer	Clinical flow cytometry laboratory director (external)	Consulting agreement CA-2027-03

2.3 Review points

Review	Date	RMF revision	Outcome
DR1 Concept	Dec 2024	A	Preliminary hazards accepted; risk controls allocated to system requirements
DR2 Architecture	Jan 2026	B	Software Class B confirmed on the basis of independent hardware controls
DR3 Verification complete	Feb 2027	C (draft)	All controls verified; 3 anomalies raised risk questions (Section 15)
DR4 Release / design transfer	May 2027	C	Overall residual risk acceptable; release approved
Annual post-market review	Dec 2028	D	P1 revised for HS-08.2; HS-17.2 added; residual risk remains acceptable

3. Intended Use and Safety Characteristics

3.1 Intended use

The CSI-100 is an automated, multi-laser flow cytometer and droplet cell sorter intended for the identification, enumeration and physical separation of cell populations in human specimens (peripheral blood, bone marrow, apheresis products and body fluids) stained with fluorochrome-labeled reagents. It is for in vitro diagnostic use by trained laboratory professionals in clinical laboratories, with IVD reagents and assays validated for the instrument. Results are used together with other clinical and laboratory findings; the instrument does not give a diagnosis.

3.2 Users, environment and patients

Operators are medical technologists with flow cytometry training (vendor course of 3 days). Laboratory supervisors approve assay templates and release results. Field service engineers are trained and certified by the manufacturer. The instrument is installed in a clinical laboratory at 15 to 30 °C, in a room with other laboratory equipment and, for infectious specimens, within the laboratory's biosafety program. Patients are not present; the patient is affected only through the result or through loss of an irreplaceable specimen.

3.3 Reasonably foreseeable misuse

- Use with reagents or assays not validated for the instrument, or reporting research-mode results as diagnostic.
- Running IVD assays after a failed or skipped QC check, or attempting to override the QC lockout.
- Opening the sort block door during sorting or before aerosol evacuation after a clog.
- Sorting infectious specimens without the aerosol management option or a biosafety enclosure.
- Editing gates or compensation in an IVD template to improve an apparently poor separation.
- Using detergents or disinfectants not listed in the IFU; filling a container with the wrong fluid.
- Service by untrained persons, including removal of laser shielding or defeating an interlock.

3.4 Safety characteristics (ISO/TR 24971 Annex A)

Question	Answer for the CSI-100	Hazards
What energy is delivered or present?	Class 3B lasers 375 to 640 nm; up to 12 kV on deflection plates; mains voltage; air up to 100 PSI; moving loading port and plate stage	HA-01, 02, 06, 09, 12
Is it used with biological materials?	Human specimens, potentially infectious; aerosol generated during droplet sorting	HA-04, HA-05
Does it produce a result used for diagnosis?	Population identification and counts; sorted fractions for molecular testing; CD34+ enumeration	HA-07, HA-08
Does it depend on software?	Yes: Class B software system (SRS-SW-CSI-010)	Section 11
Is it connected to networks?	Instrument link to workstation; optional LIS and network storage	HA-16
Are consumables or fluids used?	Sheath, bleach, ethanol, water; beads and reagents	HA-10, HA-14, HA-15
Is it installed, moved or maintained?	185 kg; installed and serviced by trained engineers	HA-11, HA-01
Could use errors cause harm?	Yes: critical tasks in Section 13	HA-17, Section 13
Could it become unavailable when needed?	QC failure or fault blocks IVD testing until corrected	HA-17 (HS-17.2)

4. Risk Analysis Methods and Criteria

4.1 Methods

Method	Scope	Record
Preliminary and system hazard analysis	Hazards and hazardous situations from intended use and safety characteristics	HA-CSI-001 Rev D (Section 6)
Design FMEA	Subsystem failure modes and their contribution to hazardous situations	DFMEA-CSI-001 Rev D (Section 12)
Process FMEA	Manufacturing and incoming inspection failure modes	PFMEA-CSI-001 Rev B (Section 12)
Fault tree analysis	Top events HS-02.3 (HV present in safe state) and HS-08.2 (drift after QC)	FTA-CSI-001 Rev B
Use FMEA, task analysis, PCA analysis	Use errors in critical tasks	HFE-CSI-001 (Section 13)
Software failure analysis	Software item contributions to hazardous situations	Section 11; SAD-SW-CSI-011 §10
Threat model (STRIDE) with CVSS 3.1	Security threats that affect safety or effectiveness	TM-SEC-CSI-001 (Section 14)

4.2 Severity and probability scales

Severity	Definition	Example for the CSI-100
S5 Catastrophic	Death	Electrocution from a live enclosure
S4 Critical	Permanent impairment or life-threatening injury; result likely to cause misdiagnosis of a serious disease	Retinal injury; laboratory-acquired HIV or HBV infection; false-negative MRD result
S3 Serious	Injury requiring medical intervention; invasive specimen re-collection	Pressure-release eye injury; repeat bone marrow aspiration
S2 Minor	Temporary injury without medical intervention; delay of result without clinical consequence	Finger pinch; bleach skin irritation; result delayed one day
S1 Negligible	Inconvenience; repeat test on the same specimen	Repeat acquisition

Level	Probability of harm P per instrument-year	Plain-language meaning (installed base of 100 instruments)
P5 Frequent	$\geq 10^{-1}$	Ten or more occurrences per year
P4 Probable	10^{-2} to $< 10^{-1}$	About one to ten per year
P3 Occasional	10^{-3} to $< 10^{-2}$	About one every one to ten years
P2 Remote	10^{-4} to $< 10^{-3}$	About one every ten to 100 years
P1 Improbable	$< 10^{-4}$	Not expected in the product life

4.3 Rules for estimating P1 and P2

- **Data first.** Estimates use, in order of preference: CSI-100 test or field data; data from the predicate instrument family (service and complaint data 2019 to 2024, PMS-PRED-001); supplier reliability data; published literature; team judgment. The basis of every estimate is written in Table 6-3.
- **Software.** Per IEC 62304 and IEC/TR 80002-1, a software failure is assumed to occur (probability 1 when the triggering condition occurs). Risk reduction is credited only to controls outside the failing software item (hardware, a separate software item on separate hardware, or an external control).

- **Use error.** Use-error rates come from formative and summative studies (HFE-CSI-001). Where 0 errors are observed in n trials, the 95% upper bound (3/n) is used for sensitivity checks.
- **External controls.** Laboratory controls (assay controls, second-person review) are credited in P2 only when required by the IFU and disclosed in labeling.
- **Rounding.** Estimates are expressed to one significant figure and rounded up when on a level boundary is uncertain.
- **Residual estimate.** A residual P1 or P2 lower than the initial estimate must be supported by effectiveness verification (Section 7).

4.4 Risk acceptability matrix

Probability \ Severity	S1	S2	S3	S4	S5
P5 Frequent	A	U	U	U	U
P4 Probable	A	R	U	U	U
P3 Occasional	A	A	R	U	U
P2 Remote	A	A	A	R	R
P1 Improbable	A	A	A	A	R

U = unacceptable (dark); R = review region, acceptable only if further reduction is not practicable and the benefit outweighs the risk (light); A = acceptable (white).

5. Hazard Identification and Harms

5.1 Hazards

Hazards were identified from the safety characteristics (Section 3.4), ISO 14971 Annex C, IEC 61010-1 and -2-101, IEC 60825-1, the predicate instrument's complaint and service history, and published biosafety guidance for cell sorters. HA-03 (splash) was merged into HA-04 in Rev B.

Hazard	Description	Hazardous situations	Worst initial	Worst residual
HA-01	Laser radiation	HS-01.1, HS-01.2, HS-01.3	Unacceptable (HS-01.1)	Acceptable (HS-01.1)
HA-02	High voltage (deflection system)	HS-02.1, HS-02.2, HS-02.3	Unacceptable (HS-02.1)	Review (benefit-risk) (HS-02.1)
HA-04	Biological: infectious specimen material and aerosol	HS-04.1, HS-04.2, HS-04.3, HS-04.4	Unacceptable (HS-04.1)	Acceptable (HS-04.1)
HA-05	Access to the sort chamber during sorting	HS-05.1	Unacceptable (HS-05.1)	Acceptable (HS-05.1)
HA-06	Moving parts	HS-06.1, HS-06.2	Acceptable (HS-06.1)	Acceptable (HS-06.1)
HA-07	Sort output not as expected	HS-07.1, HS-07.2, HS-07.3	Unacceptable (HS-07.1)	Review (benefit-risk) (HS-07.3)
HA-08	Incorrect result from configuration, QC, calibration or analysis	HS-08.1, HS-08.2, HS-08.3, HS-08.4, HS-08.5, HS-08.6	Unacceptable (HS-08.1)	Review (benefit-risk) (HS-08.1)
HA-09	Stored pressure	HS-09.1, HS-09.2	Review (benefit-risk) (HS-09.2)	Acceptable (HS-09.1)
HA-10	Chemical: decontamination agents	HS-10.1	Acceptable (HS-10.1)	Acceptable (HS-10.1)
HA-11	Instrument mass and stability	HS-11.1	Acceptable (HS-11.1)	Acceptable (HS-11.1)
HA-12	Mains electricity and fire	HS-12.1, HS-12.2	Review (benefit-risk) (HS-12.1)	Review (benefit-risk) (HS-12.1)
HA-13	Electromagnetic disturbance	HS-13.1	Review (benefit-risk) (HS-13.1)	Acceptable (HS-13.1)
HA-14	Fluid misconnection	HS-14.1	Review (benefit-risk) (HS-14.1)	Acceptable (HS-14.1)
HA-15	Fluid run-out and loss of function during sorting	HS-15.1, HS-15.2	Unacceptable (HS-15.1)	Review (benefit-risk) (HS-15.2)
HA-16	Cybersecurity	HS-16.1, HS-16.2	Unacceptable (HS-16.1)	Acceptable (HS-16.1)
HA-17	Use error in result handling; instrument availability	HS-17.1, HS-17.2	Unacceptable (HS-17.1)	Review (benefit-risk) (HS-17.1)

5.2 Harms catalogue

Each harm has one severity, set by Clinical Affairs and used for every hazardous situation that leads to it, so that the same injury is not rated differently in different parts of the analysis.

Harm	Description	Severity	Rationale
H-01	Permanent retinal injury	S4 Critical	Class 3B beam can cause irreversible retinal damage before the blink reflex (IEC 60825-1 MPE exceeded at 405 to 633 nm).
H-02	Minor skin or eye irritation from stray light	S2 Minor	Diffuse or low-power reflections; temporary, no medical intervention.

Harm	Description	Severity	Rationale
H-03	Electric shock injury from high voltage	S4 Critical	Up to 12 kV at limited current can cause burns, falls and cardiac effects; permanent injury possible.
H-04	Electrocution (death)	S5 Catastrophic	Mains voltage through the body can cause fatal cardiac arrest.
H-05	Laboratory-acquired infection	S4 Critical	Specimens may carry HIV, HBV, HCV or M. tuberculosis; infection can be life-threatening or chronic.
H-06	Pinch or minor laceration	S2 Minor	Low-force mechanisms (below IEC 61010-1 limits); first aid only.
H-07	Crush injury	S3 Serious	Instrument up to 185 kg falling on a foot or hand; fracture requiring treatment.
H-08	Chemical irritation of skin or eyes	S2 Minor	10% bleach or 70% ethanol; resolves with rinsing.
H-09	Impact or spray injury from pressure release	S3 Serious	Cap or fitting released at up to 100 PSI; eye injury requiring treatment possible.
H-10	Misdiagnosis of a hematological disease	S4 Critical	False-negative or false-positive result for leukemia, lymphoma or MRD can lead to withheld or unnecessary chemotherapy; results are interpreted with other findings, which prevents S5.
H-11	Delayed result (hours to days)	S2 Minor	Repeat testing within the clinical window; no change in treatment outcome expected for the intended use.
H-12	Invasive specimen re-collection	S3 Serious	Repeat bone marrow aspiration or apheresis; procedure risk and discomfort; medical intervention required.
H-13	Incorrect cell therapy product characterization	S4 Critical	Wrong CD34+ count could lead to inadequate graft dose and delayed engraftment; mitigated by release testing but potentially serious.
H-15	Disclosure of patient information	S2 Minor	Privacy harm; no physical injury; regulatory notification may be required.
H-16	Burns or smoke inhalation from fire	S4 Critical	Laboratory fire from instrument overheating.

H-14 (allergic reaction to latex) was retired in Rev B after latex was removed from all user-contacted parts.

6. Hazard Analysis and Risk Control

Table 6-1 estimates and evaluates the initial risk of each of the 36 hazardous situations. Table 6-2 lists the risk controls in the priority order of ISO 14971 clause 7.1 (ID = inherent safety by design, PM = protective measure in the device or manufacturing process, IS = information for safety) and the residual risk. Table 6-3 gives the basis of each estimate. Requirement IDs refer to SRS-CSI-001 Rev E (SYS) and SRS-SW-CSI-010 Rev F (RA, FW, AD, DM, CM).

Table 6-1 Risk estimation and evaluation before control

HS	Sequence of events	Hazardous situation	Harm	S	P1	P2	P	Risk
HA-01 Laser radiation								
HS-01.1	<i>Laser radiation (Class 3B, 375 to 633 nm).</i> Operator opens flow cell access door during acquisition to check the stream; shutter fails to close.	Direct or specularly reflected beam reaches the operator's eye.	H-01 Permanent retinal injury	S4	2×10^{-2}	10^{-1}	2×10^{-3} P3	U
HS-01.2	<i>Laser radiation (Class 3B).</i> Service engineer defeats the interlock to align optics with covers removed.	Service engineer exposed to open Class 3B beams.	H-01 Permanent retinal injury	S4	5×10^{-2}	5×10^{-2}	2.5×10^{-3} P3	U
HS-01.3	<i>Laser radiation (stray light).</i> Operator looks into the side-stream camera window during drop-delay setup.	Diffuse scattered 660 nm light reaches eye.	H-02 Minor skin or eye irritation from stray light	S2	10^{-1}	10^{-2}	10^{-3} P3	A
HA-02 High voltage (deflection system)								
HS-02.1	<i>High voltage on deflection plates (up to 12 kV).</i> Operator opens the sort block door to clean the aspirator while plates are energized (plates stay energized with door open).	Hand contacts an energized plate.	H-03 Electric shock injury from high voltage	S4	3×10^{-2}	3×10^{-1}	9×10^{-3} P3	U
HS-02.2	<i>High voltage.</i> Insulation degrades from fluid ingress; arc to the sort chamber door frame.	Operator touches door frame at elevated potential.	H-03 Electric shock injury from high voltage	S4	10^{-3}	3×10^{-1}	3×10^{-4} P2	R
HS-02.3	<i>High voltage.</i> Controller firmware hangs during sorting; HV stays on and warning software stops updating.	HV present while user believes system is stopped.	H-03 Electric shock injury from high voltage	S4	10^{-1}	10^{-2}	10^{-3} P3	U
HA-04 Biological: infectious specimen material and aerosol								
HS-04.1	<i>Infectious aerosol from droplet sorting.</i> Normal sorting of an infectious specimen; aerosol leaks around sort chamber seals.	Operator in the room inhales aerosol.	H-05 Laboratory-acquired infection	S4	5×10^{-1}	10^{-2}	5×10^{-3} P3	U
HS-04.2	<i>Infectious aerosol.</i> Nozzle clogs; deflected stream sprays in chamber; operator opens sort block door to clear the clog.	Operator's face near a chamber full of aerosol.	H-05 Laboratory-acquired infection	S4	2×10^{-1}	5×10^{-2}	10^{-2} P4	U
HS-04.3	<i>Infectious material (splash).</i> Tube cracks or cap leaks during loading into the pressurized sample chamber.	Specimen splashes onto operator's gloves or face.	H-05 Laboratory-acquired infection	S4	10^{-2}	10^{-2}	10^{-4} P2	R
HS-04.4	<i>Contaminated surfaces.</i> Service engineer replaces sample line on an instrument that was not decontaminated.	Contact with residual specimen material.	H-05 Laboratory-acquired infection	S4	10^{-1}	10^{-3}	10^{-4} P2	R
HA-05 Access to the sort chamber during sorting								
HS-05.1	<i>Access to sort chamber during sorting.</i> Operator opens the collection chamber door to change tubes during a sort.	Hand near energized plates and aerosol.	H-05 Laboratory-acquired infection	S4	10^{-1}	5×10^{-2}	5×10^{-3} P3	U

HS	Sequence of events	Hazardous situation	Harm	S	P1	P2	P	Risk
HA-06 Moving parts								
HS-06.1	<i>Moving loading port.</i> Operator steadies the tube while the loading port rises.	Finger trapped between port and chamber.	H-06 Pinch or minor laceration	S2	5×10^{-1}	10^{-2}	5×10^{-3} P3	A
HS-06.2	<i>Moving plate stage.</i> Operator adjusts a plate while the stage moves.	Finger caught by stage.	H-06 Pinch or minor laceration	S2	10^{-1}	10^{-2}	10^{-3} P3	A
HA-07 Sort output not as expected								
HS-07.1	<i>Low purity of sorted fraction (breakoff drift).</i> Room temperature changes during a 3 h sort; breakoff drifts; non-target cells sorted.	Sorted fraction reported pure is used for molecular testing (e.g., mutation analysis).	H-10 Misdiagnosis of a hematological disease	S4	3×10^{-1}	3×10^{-2}	9×10^{-3} P3	U
HS-07.2	<i>Wrong drop delay.</i> Drop delay set incorrectly after nozzle change.	Most sorted drops miss target cells; low purity and yield.	H-10 Misdiagnosis of a hematological disease	S4	10^{-1}	3×10^{-2}	3×10^{-3} P3	U
HS-07.3	<i>Rare-event sorting at high event rate.</i> User sorts a population below 1% at 45,000 events/s in Purity mode.	Coincident events lower purity below user expectation.	H-10 Misdiagnosis of a hematological disease	S4	10^{-1}	3×10^{-2}	3×10^{-3} P3	U
HA-08 Incorrect result from configuration, QC, calibration or analysis								
HS-08.1	<i>Optical configuration mismatch.</i> After a field upgrade, the 405 nm array keeps the 450/40 filter while the 445 nm laser is active.	Fluorescence assigned to the wrong marker; MRD population misidentified.	H-10 Misdiagnosis of a hematological disease	S4	5×10^{-2}	2×10^{-1}	10^{-2} P4	U
HS-08.2	<i>Detector sensitivity drift after a passing QC.</i> Laser power or alignment drifts during the day after morning QC passed.	Dim marker population under-detected in afternoon runs.	H-10 Misdiagnosis of a hematological disease	S4	10^{-1}	10^{-1}	10^{-2} P4	U
HS-08.3	<i>Failed or expired QC.</i> User runs an IVD assay without a passing QC in the last 24 h.	Unverified instrument generates clinical result.	H-10 Misdiagnosis of a hematological disease	S4	10^{-1}	5×10^{-2}	5×10^{-3} P3	U
HS-08.4	<i>Unapproved assay modification.</i> User edits gates in an IVD template to 'improve' separation.	Result from unvalidated analysis.	H-10 Misdiagnosis of a hematological disease	S4	10^{-1}	10^{-1}	10^{-2} P4	U
HS-08.5	<i>Compensation error.</i> Compensation calculated from poor single-stain controls.	Spillover mis-corrected; false positive population.	H-10 Misdiagnosis of a hematological disease	S4	10^{-1}	5×10^{-2}	5×10^{-3} P3	U
HS-08.6	<i>Incorrect CD34+ enumeration.</i> Low event count or wrong gating in a cell therapy product.	Graft dose miscalculated.	H-13 Incorrect cell therapy product characterization	S4	10^{-2}	10^{-1}	10^{-3} P3	U
HA-09 Stored pressure								
HS-09.1	<i>Pressurized sheath tank (up to 100 PSI supply).</i> Regulator fails open; tank pressure rises.	Tank or fitting ruptures.	H-09 Impact or spray injury from pressure release	S3	10^{-3}	10^{-1}	10^{-4} P2	A
HS-09.2	<i>Pressurized tank.</i> User opens sheath tank lid while system is pressurized.	Lid released under pressure.	H-09 Impact or spray injury from pressure release	S3	5×10^{-2}	5×10^{-2}	2.5×10^{-3} P3	R
HA-10 Chemical: decontamination agents								
HS-10.1	<i>Decontamination agents.</i> Seal degrades after repeated bleach cycles; bleach leaks under the instrument.	Operator contacts bleach.	H-08 Chemical irritation of skin or eyes	S2	10^{-2}	10^{-1}	10^{-3} P3	A
HA-11 Instrument mass and stability								
HS-11.1	<i>Instrument mass (185 kg).</i> Instrument moved on an unsuitable bench.	Instrument tips.	H-07 Crush injury	S3	10^{-3}	3×10^{-1}	3×10^{-4} P2	A

HS	Sequence of events	Hazardous situation	Harm	S	P1	P2	P	Risk
HA-12 Mains electricity and fire								
HS-12.1	<i>Mains voltage.</i> Protective earth broken in power cord; insulation fault.	Enclosure becomes live.	H-04 Electrocution (death)	S5	10^{-3}	10^{-1}	10^{-4} P2	R
HS-12.2	<i>Overheating electronics.</i> Fan failure; power supply overheats.	Component ignition.	H-16 Burns or smoke inhalation from fire	S4	10^{-3}	10^{-2}	10^{-5} P1	A
HA-13 Electromagnetic disturbance								
HS-13.1	<i>Electromagnetic disturbance.</i> Nearby equipment radiates 3 V/m at 80 to 120 MHz.	Noise events alter population percentages.	H-10 Misdiagnosis of a hematological disease	S4	10^{-2}	5×10^{-2}	5×10^{-4} P2	R
HA-14 Fluid misconnection								
HS-14.1	<i>Fluid misconnection.</i> Bleach container connected to sheath port.	Cells lysed during sort; specimen lost.	H-12 Invasive specimen re-collection	S3	2×10^{-2}	3×10^{-1}	6×10^{-3} P3	R
HA-15 Fluid run-out and loss of function during sorting								
HS-15.1	<i>Fluid run-out.</i> Sheath empties during a 4 h sort.	Sort stops; sample partly consumed.	H-12 Invasive specimen re-collection	S3	10^{-1}	2×10^{-1}	2×10^{-2} P4	U
HS-15.2	<i>Loss of function during sort.</i> Clog or software fault stops a sort of an irreplaceable specimen.	Sample consumed without complete result.	H-12 Invasive specimen re-collection	S3	2×10^{-1}	5×10^{-2}	10^{-2} P4	U
HA-16 Cybersecurity								
HS-16.1	<i>Unauthorized modification.</i> Attacker on the lab network replays commands or installs modified software.	Results or settings altered undetected.	H-10 Misdiagnosis of a hematological disease	S4	10^{-2}	10^{-1}	10^{-3} P3	U
HS-16.2	<i>Data exposure.</i> Workstation stolen or exported data shared.	Patient data disclosed.	H-15 Disclosure of patient information	S2	10^{-2}	5×10^{-1}	5×10^{-3} P3	A
HA-17 Use error in result handling; instrument availability								
HS-17.1	<i>Specimen misidentification.</i> Operator types wrong specimen ID; result released to wrong patient.	Wrong result in patient record.	H-10 Misdiagnosis of a hematological disease	S4	2×10^{-2}	3×10^{-1}	6×10^{-3} P3	U
HS-17.2	<i>Instrument unavailable.</i> QC fails repeatedly (e.g., thermal drift); IVD testing blocked until service.	Result delayed by one or more days.	H-11 Delayed result (hours to days)	S2	5×10^{-2}	10^{-1}	5×10^{-3} P3	A

U = unacceptable, R = review region, A = acceptable (Section 4.4).

Table 6-2 Risk controls and residual risk

HS	Initial	Risk controls (type, measure, requirement)	P1	P2	P	Residual	Basis for residual estimate
HA-01 Laser radiation							
HS-01.1	S4 P3 U	ID Enclosed beam path; Class 1 in all user-accessible conditions (SYS-511) PM Hardware shutter blocks all beams when door opens; firmware rejects conflicting commands (SYS-512, FW-502) IS Warning labels; IFU (SYS-513, SYS-1103)	10^{-5}	10^{-1}	10^{-6} P1	S4 P1 A	Shutter is a normally-closed hardware interlock (single-fault safe per IEC 60825-1); 59 of 59 door cycles blocked the beam.
HS-01.2	S4 P3 U	ID Factory-fixed alignment; no routine field alignment of launch plate (SYS-436) PM Service override requires Admin Application service role and key; auto-expires after 30 min (AD-901, CM-102) IS Service manual laser safety procedure, eyewear by wavelength, training (SYS-1104)	5×10^{-3}	10^{-2}	5×10^{-5} P1	S4 P1 A	Field alignment reduced to launch plate replacement only (~0.1 per instrument-year); training records required.
HS-01.3	S2 P3 A	ID Camera window with laser-line filter (SYS-511) IS IFU (SYS-1102)	10^{-1}	10^{-4}	10^{-5} P1	S2 P1 A	Filtered window keeps exposure below Class 1 AEL.
HA-02 High voltage (deflection system)							
HS-02.1	S4 P3 U	ID Plates recessed and insulated; no energized part accessible with door closed (SYS-523) PM Red warning light and on-screen indicator; HV removed by e-stop (SYS-522, SYS-532, RA-1003) IS HV label at sort block; IFU cleaning procedure requires HV off (SYS-1103)	3×10^{-3}	5×10^{-2}	1.5×10^{-4} P2	S4 P2 R	Formative and summative testing: 0 of 30 users opened with HV indicator on after training; plate recess limits contact.
HS-02.2	S4 P2 R	ID Creepage and clearance per IEC 61010-1; sealed HV feedthroughs; drip channels (SYS-501, SYS-523) PM HV readback fault triggers safe state; 100% hipot in production (FW-307, FW-606, SYS-502)	10^{-5}	3×10^{-1}	3×10^{-6} P1	S4 P1 A	Design and 100% production hipot; readback detects leakage-induced drop.
HS-02.3	S4 P3 U	PM Independent safety supervisor removes HV enable within 250 ms of missing heartbeat (FW-604, FW-605) PM Hardware warning light driven from HV supply, not from software (SYS-522)	10^{-3}	10^{-2}	10^{-5} P1	S4 P1 A	Two independent channels (SI-15 and hardwired lamp); fault injection 100% effective.
HA-04 Biological: infectious specimen material and aerosol							
HS-04.1	S4 P3 U	ID Sealed sort block and collection chamber (SYS-804) PM Aerosol evacuation module with ULPA filter, ≥ 7 CFM (SYS-551, SYS-552) IS IFU biosafety chapter: risk assessment, BSC or evacuation for infectious specimens (SYS-554)	5×10^{-1}	10^{-4}	5×10^{-5} P1	S4 P1 A	Containment test: no challenge particles detected outside chamber; P2 reduced by evacuation and facility biosafety.
HS-04.2	S4 P4 U	PM Clog detection < 500 ms; stream and HV off, drawer closed, tube unloaded within 2 s (SYS-543, FW-210, FW-602) PM Guided recovery: boost evacuation before door-open step is shown (RA-504, RA-505) IS IFU clog handling (SYS-554)	2×10^{-3}	10^{-2}	2×10^{-5} P1	S4 P1 A	Post-market logs: 100% of 212 clog recoveries used boost evacuation (Section 16.2).

HS	Initial	Risk controls (type, measure, requirement)	P1	P2	P	Residual	Basis for residual estimate
HS-04.3	S4 P2 R	ID Splash shield and chamber cover; pressurization only after port closed (RA-205) IS IFU: approved tube types only; PPE (SYS-1102)	10 ⁻³	10 ⁻³	10 ⁻⁶ P1	S4 P1 A	Interlocked pressurization; splash contained.
HS-04.4	S4 P2 R	PM Automated bleach clean cycle (SYS-603, RA-203) IS Decontamination certificate required before service (SYS-554)	10 ⁻²	10 ⁻³	10 ⁻⁵ P1	S4 P1 A	Service procedure requires certificate; clean cycle validated.
HA-05 Access to the sort chamber during sorting							
HS-05.1	S4 P3 U	ID Collection door cannot open unless sort block door is closed (SYS-521) PM Firmware door monitoring (FW-502) IS IFU tube change (SYS-1102)	10 ⁻⁴	5×10 ⁻²	5×10 ⁻⁶ P1	S4 P1 A	Mechanical interlock; 59 of 59 attempts blocked.
HA-06 Moving parts							
HS-06.1	S2 P3 A	ID Force below IEC 61010-1 pinch limit (SYS-562) PM Illuminated e-stop; stall detection (SYS-531, SYS-533, FW-309) IS Pinch label (SYS-1103)	5×10 ⁻¹	10 ⁻⁴	5×10 ⁻⁵ P1	S2 P1 A	Force measured 38 N (limit 50 N); stall reverses motor.
HS-06.2	S2 P3 A	PM Collection chamber door must be closed for stage motion; e-stop stops stage in 100 ms (SYS-532, FW-601) IS IFU (SYS-1102)	10 ⁻³	10 ⁻²	10 ⁻⁵ P1	S2 P1 A	Door interlock prevents access during motion.
HA-07 Sort output not as expected							
HS-07.1	S4 P3 U	PM Breakoff control; charging paused within 100 ms when out of range (SYS-613, SYS-542, FW-205) PM Sort report with pauses and conflicts (RA-706) IS IFU: re-analyze sorted fraction (SYS-1102)	3×10 ⁻³	3×10 ⁻²	9×10 ⁻⁵ P1	S4 P1 A	Induced drift: charging stopped in 100% of trials; purity maintained > 98%.
HS-07.2	S4 P3 U	PM Automatic drop delay with beads (SYS-465, RA-407) PM Nozzle confirmation loads matching setup (RA-402)	10 ⁻³	3×10 ⁻²	3×10 ⁻⁵ P1	S4 P1 A	Auto delay within ±0.25 drop of manual in all tests.
HS-07.3	S4 P3 U	PM Software advisory when sort population < 1% (SYS-633) IS IFU event rate guidance (SYS-1102)	10 ⁻²	3×10 ⁻²	3×10 ⁻⁴ P2	S4 P2 R	Advisory shown in 100% of test cases; VAL-CSI-02 supplementary specimens met 95%.
HA-08 Incorrect result from configuration, QC, calibration or analysis							
HS-08.1	S4 P4 U	ID Configuration generated from controlled record with rules (AD-105, AD-106) PM Board inventory and configuration match check before acquisition (FW-302, RA-805) PM New baseline required after change (AD-104)	5×10 ⁻⁴	2×10 ⁻¹	10 ⁻⁴ P2	S4 P2 R	Seeded configuration errors blocked in 100% of STS cases (ANM-0026 closed).
HS-08.2	S4 P4 U	PM Daily QC lockout (SYS-622, RA-803) PM Assay controls with every run; results flagged when warnings active (RA-903) PM Fan control on laser temperature; laser temperature warning (added Rev D) (FW-311, RA-107) PM Thermal margin requirement (added Rev D) (SYS-327)	10 ⁻³	10 ⁻¹	10 ⁻⁴ P2	S4 P2 R	After CAPA-2028-014: 0 drift events in 5,412 QC runs (90 days); laser temp 99th percentile 30.4 °C.
HS-08.3	S4 P3 U	PM QC lockout blocks IVD templates; UTC-based 24 h window (RA-803) PM Override limited to research use and labeled (RA-804)	10 ⁻⁵	5×10 ⁻²	5×10 ⁻⁷ P1	S4 P1 A	Lockout cannot be bypassed for IVD templates (STS, ANM-0025).

HS	Initial	Risk controls (type, measure, requirement)	P1	P2	P	Residual	Basis for residual estimate
HS-08.4	S4 P4 U	PM Locked, signed assay templates; versioning (SYS-704, RA-901, AD-802) PM Audit trail (DM-201)	10 ⁻⁴	10 ⁻¹	10 ⁻⁵ P1	S4 P1 A	Operators cannot edit locked templates (STS).
HS-08.5	S4 P3 U	PM Compensation locked in IVD templates (RA-901) IS IFU and assay IFU (SYS-1102)	10 ⁻³	5×10 ⁻²	5×10 ⁻⁵ P1	S4 P1 A	Compensation part of approved template.
HS-08.6	S4 P3 U	PM Locked CD34 template with minimum event count (RA-901) PM Acquisition rate and lost events reported (RA-304)	10 ⁻³	10 ⁻¹	10 ⁻⁴ P2	S4 P2 R	VAL-CSI-01 CD34 agreement within ±10%.
HA-09 Stored pressure							
HS-09.1	S3 P2 A	ID Tanks rated 1.5× maximum with relief valves (SYS-561) PM Pressure regulation and monitoring (FW-401)	10 ⁻⁶	10 ⁻¹	10 ⁻⁷ P1	S3 P1 A	Relief valve is mechanical and independent.
HS-09.2	S3 P3 R	PM Venting on power off; refill procedure depressurizes (SYS-524, FW-403) IS Tank label; IFU (SYS-1102)	5×10 ⁻³	10 ⁻²	5×10 ⁻⁵ P1	S3 P1 A	Vent within 10 s verified.
HA-10 Chemical: decontamination agents							
HS-10.1	S2 P3 A	ID Compatible materials; drip tray (SYS-553)	10 ⁻⁴	10 ⁻¹	10 ⁻⁵ P1	S2 P1 A	500-cycle compatibility test pass.
HA-11 Instrument mass and stability							
HS-11.1	S3 P2 A	ID Stable at 10° tilt (SYS-563) IS Installation by service only; site preparation guide (SYS-1104, SYS-1203)	10 ⁻⁴	3×10 ⁻¹	3×10 ⁻⁵ P1	S3 P1 A	Tilt test pass.
HA-12 Mains electricity and fire							
HS-12.1	S5 P2 R	ID IEC 61010-1 double protection (SYS-501) PM 100% ground bond, hipot, leakage in production (SYS-502) PM Circuit breaker (SYS-315)	10 ⁻⁶	10 ⁻¹	10 ⁻⁷ P1	S5 P1 R	Two independent means of protection.
HS-12.2	S4 P1 A	ID Flame-rated enclosure and materials (SYS-501) PM Temperature monitoring to safe state (FW-308, FW-606)	10 ⁻⁴	10 ⁻³	10 ⁻⁷ P1	S4 P1 A	Fault test with blocked fans.
HA-13 Electromagnetic disturbance							
HS-13.1	S4 P2 R	ID Shielded detector cabling; filters (SYS-503) PM Daily QC (SYS-621)	10 ⁻⁴	5×10 ⁻²	5×10 ⁻⁶ P1	S4 P1 A	IEC 61326-2-6 pass with performance criteria (ANM-0021).
HA-14 Fluid misconnection							
HS-14.1	S3 P3 R	ID Keyed non-interchangeable connectors; color coding (SYS-801, SYS-905)	10 ⁻⁵	3×10 ⁻¹	3×10 ⁻⁶ P1	S3 P1 A	Misconnection physically impossible (tested).
HA-15 Fluid run-out and loss of function during sorting							
HS-15.1	S3 P4 U	PM Level monitoring; capacity check before sort (SYS-493, RA-207, FW-402)	10 ⁻³	2×10 ⁻¹	2×10 ⁻⁴ P2	S3 P2 A	Sort blocked when capacity insufficient.
HS-15.2	S3 P4 U	PM Sample tube unloaded on clog (sample preserved) (FW-602) PM Sort record preserved; resume (RA-703, DM-104)	2×10 ⁻¹	5×10 ⁻³	10 ⁻³ P3	S3 P3 R	Sample returned to tube in all clog tests.

HS	Initial	Risk controls (type, measure, requirement)	P1	P2	P	Residual	Basis for residual estimate
HA-16 Cybersecurity							
HS-16.1	S4 P3 U	ID Isolated instrument link; locked image (SYS-707, CM-201) PM Signed updates; pairing; session keys (FW-104, FW-802, FW-803) PM Audit trail (DM-201)	10^{-5}	10^{-1}	10^{-6} P1	S4 P1 A	Penetration test after ANM-0030: no exploitable path.
HS-16.2	S2 P3 A	PM Encryption at rest; access control (DM-504, CM-101) PM De-identified export option (DM-403)	10^{-2}	10^{-3}	10^{-5} P1	S2 P1 A	Encrypted database unreadable without credentials.
HA-17 Use error in result handling; instrument availability							
HS-17.1	S4 P3 U	PM Review and release with e-signature (DM-301, SYS-703) PM LIS export avoids transcription (DM-404) IS IFU and training (SYS-1102)	10^{-3}	10^{-1}	10^{-4} P2	S4 P2 R	LIS export used at 70% of sites removes transcription; ID confirmation at release. P2 credits second-person review against clinical history (laboratory procedure, disclosed). Instrument cannot fully prevent entry error; barcode entry planned.
HS-17.2	S2 P3 A	PM Fan control and thermal margin (CAPA-2028-014) (FW-311, SYS-327) PM Laser temperature visible for faster troubleshooting (RA-107)	5×10^{-3}	10^{-1}	5×10^{-4} P2	S2 P2 A	QC failures from thermal cause 0 in 90 days post-CAPA.

Table 6-3 Basis of probability estimates

HS	P1 basis (initial)	P2 basis (initial)
HA-01 Laser radiation		
HS-01.1	2×10 ⁻² : Door opened ~5 times per day (task analysis); without a shutter every opening exposes the beam path.	10 ⁻¹ : Beam is focused in the flow cell; eye alignment with the reflected path is unlikely but possible (optics analysis).
HS-01.2	5×10 ⁻² : Alignment work in the field ~1 per instrument-year (service plan).	5×10 ⁻² : Trained engineers; eyewear not always worn (industry incident reports).
HS-01.3	10 ⁻¹ : Drop-delay setup daily; window viewing plausible.	10 ⁻² : Diffuse, low-power light; below MPE in most geometries.
HA-02 High voltage (deflection system)		
HS-02.1	3×10 ⁻² : Aspirator cleaning weekly; plates left on in ~5% of cases (formative study).	3×10 ⁻¹ : Contact likely once hand is inside the sort block.
HS-02.2	10 ⁻³ : DFMEA: insulation breakdown from saline ingress without design controls.	3×10 ⁻¹ : Door frame touched each tube change.
HS-02.3	10 ⁻¹ : Software failure probability taken as 1 per IEC/TR 80002-1; hang scenarios in FMEA ~0.1 per year.	10 ⁻² : Requires user to open door and reach plates.
HA-04 Biological: infectious specimen material and aerosol		
HS-04.1	5×10 ⁻¹ : Laboratories sort potentially infectious specimens routinely; aerosol always present in chamber.	10 ⁻² : Infection needs viable agent and dose (biosafety literature for cell sorters).
HS-04.2	2×10 ⁻¹ : Clogs ~1 per instrument-month (predicate experience); door opened quickly in most cases without guidance.	5×10 ⁻² : High aerosol concentration after clog.
HS-04.3	10 ⁻² : Tube failure under pressure (predicate complaint data).	10 ⁻² : Mucosal exposure needed.
HS-04.4	10 ⁻¹ : Sample line replacement every 4 to 6 months.	10 ⁻³ : Gloves worn; low residual volume.
HA-05 Access to the sort chamber during sorting		
HS-05.1	10 ⁻¹ : Tube changes during long sorts common.	5×10 ⁻² : Aerosol present during sort.
HA-06 Moving parts		
HS-06.1	5×10 ⁻¹ : About 40 tube loads per day, 250 days (10,000 loads per instrument-year). Formative study: operators steadied the tube with a finger in 2 of 30 sessions; a finger in the gap at the moment of closure estimated at 1 in 20,000 loads.	10 ⁻² : Closing force at the gap is low; a trapped finger is expected to cause a pinch without injury in most cases (IEC 61010-1 Annex on pinch points).
HS-06.2	10 ⁻¹ : Plate sorting at about 30% of sites, weekly; adjusting a plate during motion observed once in formative testing.	10 ⁻² : Stage speed 50 mm/s and drive force limited; contact results in a minor pinch only in about 1 in 100 contacts.
HA-07 Sort output not as expected		
HS-07.1	3×10 ⁻¹ : Predicate data: breakoff drift beyond ±1 drop in about 30% of sorts longer than 2 h without active breakoff control.	3×10 ⁻² : Contamination of a few percent rarely changes a molecular result; clinically significant change estimated at 3 in 100 affected sorts.
HS-07.2	10 ⁻¹ : Nozzle changes about 50 per instrument-year; manual drop-delay errors in 1 of 5 training sessions; an unchecked error reaching a clinical sort about 1 in 500 changes.	3×10 ⁻² : A wrong drop delay usually gives very low yield, which users notice; a purity error that reaches a report is less common.
HS-07.3	10 ⁻¹ : Clinical sorts of populations below 1% at above 40,000 events/s expected at least once per 10 instrument-years (beta site logs; ANM-0056).	3×10 ⁻² : Purity loss is moderate (to about 90%); downstream tests are affected in a minority of cases.

HS	P1 basis (initial)	P2 basis (initial)
HA-08 Incorrect result from configuration, QC, calibration or analysis		
HS-08.1	5×10 ⁻² : Field upgrades about 0.2 per instrument-year; filter rule missed in 1 of 4 upgrades without a rule check (service review of the predicate).	2×10 ⁻¹ : Assay controls use fixed channels and may not detect a spectral shift in every channel.
HS-08.2	10 ⁻¹ : Rev C estimate 1×10 ⁻² from 1,000 h laser stability testing. Revised in Rev D from post-market data: 7 of 64 instruments drifted in 2028 (0.11 per instrument-year; CAPA-2028-014).	10 ⁻¹ : Assay controls with each run detect most drifts.
HS-08.3	10 ⁻¹ : Laboratory surveys (LIT-CSI-2028-01) report skipped QC where no lockout exists; an IVD run without valid QC assumed at least once per 10 instrument-years.	5×10 ⁻² : The instrument is usually still within specification when QC is skipped.
HS-08.4	10 ⁻¹ : Template edits to 'improve' separation reported in manual workflows; predicate audit trails show edits in about 10% of instrument-years.	10 ⁻¹ : An edit changes population boundaries enough to change a clinical result in about 1 case in 10.
HS-08.5	10 ⁻¹ : Compensation recalculated from poor controls in about 10% of instrument-years (operator-dependent; predicate service data).	5×10 ⁻² : Most compensation errors are visible to experienced reviewers.
HS-08.6	10 ⁻² : CD34+ enumeration performed at about 10% of sites; low event count or gating error in about 1 in 10 of those instrument-years.	10 ⁻¹ : Release testing of cell therapy products includes independent viability and count checks.
HA-09 Stored pressure		
HS-09.1	10 ⁻³ : Regulator failure rate 1×10 ⁻³ per year (supplier reliability data RD-V14-02).	10 ⁻¹ : Failure must also rupture a fitting while a person is within 1 m.
HS-09.2	5×10 ⁻² : Sheath refill 3 times per week; lid opened while pressurized in about 1 in 3,000 refills (formative study).	5×10 ⁻² : Pressure at the lid with stream off is below 10 PSI in most cases.
HA-10 Chemical: decontamination agents		
HS-10.1	10 ⁻² : Seal aging under repeated bleach exposure: 1 leak per 100 instrument-years expected (predicate service data).	10 ⁻¹ : Leaks are usually noticed on the drip tray before contact.
HA-11 Instrument mass and stability		
HS-11.1	10 ⁻³ : Relocation by users about once per 10 instrument-years; unsuitable bench in 1 of 100 relocations.	3×10 ⁻¹ : A tipping instrument injures a person nearby in about 3 of 10 cases.
HA-12 Mains electricity and fire		
HS-12.1	10 ⁻³ : Single fault (broken protective earth) plus insulation fault: 1×10 ⁻³ per year (IEC 61010-1 analysis).	10 ⁻¹ : The enclosure is touched daily, so contact is likely once it is live.
HS-12.2	10 ⁻³ : Fan failure 1×10 ⁻³ per year (supplier MTBF 70,000 h, derated).	10 ⁻² : Flame-rated enclosure contains most ignition events.
HA-13 Electromagnetic disturbance		
HS-13.1	10 ⁻² : Radiated disturbance of 3 V/m in the 80 to 120 MHz band in clinical labs is uncommon (site surveys, predicate).	5×10 ⁻² : QC and assay controls detect most noise-related shifts.
HA-14 Fluid misconnection		
HS-14.1	2×10 ⁻² : Without keyed connectors, container swaps were seen in 2 of 100 refill observations (formative); a swap that reaches a running sort is estimated at 1 per 50 instrument-years.	3×10 ⁻¹ : The specimen may be irreplaceable (bone marrow, apheresis).
HA-15 Fluid run-out and loss of function during sorting		
HS-15.1	10 ⁻¹ : Sorts longer than 2 h are routine; sheath not refilled before a long sort about once per 10 instrument-years (predicate service data).	2×10 ⁻¹ : The remaining sample may be insufficient for repeat testing, requiring re-collection.

HS	P1 basis (initial)	P2 basis (initial)
HS-15.2	2×10^{-1} : About 12 clogs per instrument-year, 2% during irreplaceable specimens (predicate data).	5×10^{-2} : Most interrupted sorts can be resumed with the remaining sample.
HA-16 Cybersecurity		
HS-16.1	10^{-2} : Threat model: exploitability medium without controls (CVSS 7.5, TH-01).	10^{-1} : Altered results may be caught by QC, assay controls or review.
HS-16.2	10^{-2} : Workstation theft or loss about 1 per 100 instrument-years (hospital IT incident data).	5×10^{-1} : Data are readable without encryption.
HA-17 Use error in result handling; instrument availability		
HS-17.1	2×10^{-2} : Manual entry error rate ~1 in 1,000 entries (literature on transcription).	3×10^{-1} : Error may be detected by clinical inconsistency.
HS-17.2	5×10^{-2} : ADDED Rev D from post-market data (CAPA-2028-014): QC-blocked days observed.	10^{-1} : Most results not time-critical within one day.

7. Verification of Risk Controls

Each risk control is verified for implementation (the control is present as specified) and for effectiveness (the control reduces P1 or P2 as claimed in Table 6-2). Verification is planned in VVP-CSI-001 Section 8 and traced in TM-CSI-DT-01 Matrix D. Controls added in Rev D were verified under CAPA-2028-014 and design change DCR-2028-031.

HS	Implementation verification	Effectiveness verification	Result
HS-01.1	DVP-CSI-502; SVR-CSI-001 (IEC 60825-1)	Accessible emission below Class 1 AEL with door open (all builds); shutter fault injection	Pass
HS-01.2	SM-CSI-001 review; TRN-CSI-003	Service validation at beta sites (VAL-CSI-10): procedure followed in 100% of observed interventions	Pass
HS-01.3	SVR-CSI-001	Emission measurement at window	Pass
HS-02.1	DVP-CSI-503	VAL-CSI-03 critical task; pinch and reach test with standard test finger	Pass
HS-02.2	SVR-CSI-001; FTS-CSI-001 §5.16	Fluid ingress test with saline; fault injection	Pass
HS-02.3	ITS-CSI-010 §10	Fault injection: SI-10 hang, SI-11 hang, bus fault (59 trials each)	Pass
HS-04.1	DVP-CSI-504	Fluorescent microsphere aerosol challenge (3 conditions × 3 runs)	Pass
HS-04.2	DVP-CSI-503 (59 trials)	DVP-CSI-504 simulated clog; VAL-CSI-03; ANM-0011, ANM-0051	Pass
HS-04.3	DVP-CSI-408	Seeded cracked-tube test	Pass
HS-04.4	DVP-CSI-504	Residual protein test after clean cycle	Pass
HS-05.1	DVP-CSI-503	Door sequence test	Pass
HS-06.1	DVP-CSI-505	Force measurement; stall test	Pass
HS-06.2	DVP-CSI-503	E-stop timing	Pass
HS-07.1	ITS-CSI-010 §11	DVP-CSI-405 induced drift; VAL-CSI-02	Pass
HS-07.2	DVP-CSI-405	VAL-CSI-02	Pass
HS-07.3	STS-CSI-010 §A7	VAL-CSI-02 supplementary	Pass
HS-08.1	STS-CSI-010 §C1	Seeded-error test; CAPA-2028-014 PA11 inventory	Pass
HS-08.2	CAPA-2028-014 CA1, CA7	CAPA effectiveness check E1 to E3	Pass
HS-08.3	STS-CSI-010 §A8	Seeded tests incl. DST boundary	Pass
HS-08.4	STS-CSI-010 §A, §D	VAL-CSI-09	Pass
HS-08.5	STS-CSI-010 §A	VAL-CSI-01	Pass
HS-08.6	STS-CSI-010 §A	VAL-CSI-01, VAL-CSI-04	Pass
HS-09.1	DVP-CSI-505	Relief valve test	Pass
HS-09.2	DVP-CSI-505	Depressurization test	Pass
HS-10.1	DVP-CSI-504	Material test	Pass
HS-11.1	DVP-CSI-505	Tilt test	Pass
HS-12.1	SVR-CSI-001	IEC 61010 tests	Pass
HS-12.2	SVR-CSI-001	Abnormal operation tests	Pass
HS-13.1	SVR-CSI-001	Immunity with monitoring	Pass

HS	Implementation verification	Effectiveness verification	Result
HS-14.1	DVP-CSI-801	Misconnection attempts	Pass
HS-15.1	DVP-CSI-408	Low-level scenarios	Pass
HS-15.2	DVP-CSI-503	Clog test	Pass
HS-16.1	DVP-CSI-701	Penetration test	Pass
HS-16.2	STS-CSI-010 §D	Security test	Pass
HS-17.1	STS-CSI-010 §D	VAL-CSI-03, VAL-CSI-09	Pass
HS-17.2	CAPA-2028-014	Effectiveness check E1	Pass

8. Worked Examples of Risk Estimation

Two hazardous situations are worked through in full to show how the estimates in Section 6 were produced and how they are checked against data. The same method was applied to every hazardous situation; the working for all others is held in HA-CSI-001 Rev D worksheets.

8.1 HS-04.2 Operator exposed to aerosol after a nozzle clog

Sequence of events. A cell clump blocks the 70 μm nozzle during a sort of a potentially infectious specimen. The deflected stream hits the sort block walls and generates a dense aerosol. The operator opens the sort block door to clear the clog before the aerosol has been removed and is exposed at face height.

Initial P1. Predicate data (PMS-PRED-001) show about 12 clogs per instrument-year. In the formative study FS-CSI-02, 18 of 30 operators (0.6) opened the door within 20 s of a clog when no guidance was given. Laboratory surveys show about 3% of sorts involve specimens known or suspected to be infectious. $P1 = 12 \times 0.6 \times 0.03 = 0.22$, entered as 2×10^{-1} .

Initial P2. 5×10^{-2} . Aerosol concentration after a clog is 10 to 100 times that of normal sorting (DVP-CSI-504 baseline); infection still requires a viable agent in a sufficient dose and a failure of personal protective equipment. Biosafety literature for cell sorters supports a conditional probability of a few percent.

Initial risk. $P = 10^{-2}$, level P4; severity S4 (laboratory-acquired infection); **unacceptable**.

Controls. (1) Firmware detects the clog within 500 ms and, without the workstation, stops the stream, removes deflection HV, closes the aspirator drawer and unloads the sample tube within 2 s (SYS-543, FW-210, FW-602). (2) The Research Application guides recovery and shows the door-open step only after boost evacuation has run (RA-504, RA-505). (3) The IFU describes clog handling (SYS-554). Summative testing after ANM-0051 found 0 of 30 operators opened the door early.

Residual P1. The fraction of clogs followed by early door opening is estimated at 0.5% (the guided step is withheld until evacuation completes; the door is not locked). $P1 = 12 \times 0.005 \times 0.03 = 1.8 \times 10^{-3}$, entered as 2×10^{-3} .

Sensitivity check: with 0 early openings in 242 observations (30 summative plus 212 logged field recoveries), the 95% upper bound is $3/242 = 1.2\%$, giving $P1 = 4.5 \times 10^{-3}$ and $P = 4.5 \times 10^{-5}$, still level P1.

Residual P2. Boost evacuation lowers the chamber particle count by more than 95% before the door-open step (DVP-CSI-504 simulated clog), so P2 falls to 10^{-2} .

Residual risk. $P = 2 \times 10^{-5}$, level P1; S4; **acceptable**. Field data for 2028 (Section 21) confirm the estimate.

8.2 HS-08.2 Sensitivity drift after a passing QC (revised in Rev D)

Sequence of events. Morning QC passes. During the day the violet laser base temperature rises, the laser output falls and dim violet-excited populations (for example CD34 dim or CD38 dim MRD populations) are under-detected in afternoon runs.

Rev C estimate. $P1 = 1 \times 10^{-2}$ from 1,000 h laser stability testing at 25 °C (one power excursion beyond the QC limit per 100 instrument-years expected). With $P2 = 1 \times 10^{-1}$, $P = 1 \times 10^{-3}$ (P3, unacceptable before control); with QC lockout and assay controls the residual was placed at P2 (review region).

What the field data showed. In 2028, 7 of 64 instruments showed thermal drift (CAPA-2028-014). The cause was a supplier TEC firmware change combined with fan control driven only by room temperature. The observed rate, 0.11 per instrument-year, is ten times the Rev C estimate. P1 is revised to 10^{-1} , which moves the initial risk to $P = 10^{-2}$, level P4. The existing controls worked: the QC lockout and assay controls caught every event, no released result differed from its repeat beyond precision limits, and no complaint alleged a wrong result.

Rev D controls. Fan control on laser base temperature (FW-311), a laser temperature warning in the Research Application (RA-107), a thermal margin requirement for all laser modules at 30 °C room temperature (SYS-327), incoming hot test and firmware check for lasers, and a 6 h thermal soak in factory acceptance testing (PFMEA, Section 12).

Residual estimate and its limits. Residual $P1 = 10^{-3}$ rests on design verification: at 30 °C room temperature, laser base temperature stayed at or below 30.4 °C (99th percentile) across 3 laser lots, which leaves 4.6 °C margin to the power-drop threshold (CA7). Field data after the update (0 drift events in 5,412 QC runs over 90 days) are consistent with this, but 90 days on 64 instruments (about 16 instrument-years) cannot on their own show a rate below 10^{-1} . The residual is therefore accepted on the design evidence, and the estimate will be re-checked at 12 months against at least 60 instrument-years of data (PMS action PM-2029-02).

Residual risk. $P = 10^{-4}$, level P2; S4; **review (benefit-risk)**, justified in Section 18.

9. Risk Profile Before and After Control

Before risk control (number of hazardous situations in each cell):

Probability \ Severity	S1	S2	S3	S4	S5
P5 Frequent	A	U	U	U	U
P4 Probable	A	R	U (2)	U (4)	U
P3 Occasional	A	A (6)	R (2)	U (14)	U
P2 Remote	A	A	A (2)	R (4)	R (1)
P1 Improbable	A	A	A	A (1)	R

After risk control (residual risk):

Probability \ Severity	S1	S2	S3	S4	S5
P5 Frequent	A	U	U	U	U
P4 Probable	A	R	U	U	U
P3 Occasional	A	A	R (1)	U	U
P2 Remote	A	A (1)	A (1)	R (6)	R
P1 Improbable	A	A (5)	A (4)	A (17)	R (1)

Of 36 hazardous situations, 20 were unacceptable and 7 were in the review region before control. After control none is unacceptable, 8 are in the review region (HS-02.1, HS-07.3, HS-08.1, HS-08.2, HS-08.6, HS-12.1, HS-15.2, HS-17.1) and 28 are acceptable. The review-region risks are justified in Section 18.

10. Risks Arising from Risk Controls and Completeness

10.1 Risks arising from risk controls (clause 7.5)

Risk control	Possible new hazard or effect	Evaluation	Outcome
QC lockout (RA-803)	IVD testing blocked; result delayed	Not analyzed before Rev D. 41 QC-blocked events in 2028.	New HS-17.2 (acceptable)
Clog response unloads tube and stops sort (FW-602)	Sort interrupted; specimen partly consumed	Specimen returned to tube; resume function	Covered by HS-15.2
Laser shutter on door open (SYS-512)	Acquisition interrupted	Data up to the event saved; no harm	No new hazard
Fan control on laser temperature (FW-311)	Higher acoustic noise; faster dust loading	Measured 60 dBA at 1 m at maximum fan speed, within the SRS §3.4 limit; filter interval unchanged	No new hazard
Account lockout after failed logins (CM-101)	Urgent result delayed	Supervisor unlock within minutes; S2, P2	Acceptable; no new HS
Encryption at rest (DM-504)	Data unreadable if key lost	Key escrow in backup procedure; restore test passed	No new hazard
Collection door interlock (SYS-521)	Tube change needs sort pause; users may be tempted to defeat interlock	Interlock not defeatable without tools; pause function provided	No new hazard
Service override expiry 30 min (AD-901)	Shutter closes during alignment	Fails to the safe condition	No new hazard
Signed software updates (FW-104)	Security patches delayed by validation	Patch process target 30 days (Section 14)	Covered by HS-16.1

10.2 Completeness of risk control (clause 7.6)

All 36 hazardous situations have been evaluated, each has at least one verified risk control where the initial risk was not acceptable, and every control is traced to a requirement and a verification record. The completeness check at this revision found 0 control requirements without a matching requirement in SRS-CSI-001 or SRS-SW-CSI-010; the five requirements added by CAPA-2028-014 (FW-311, FW-312, FW-313, RA-107, SYS-327) are included in SRS-CSI-001 Rev E and SRS-SW-CSI-010 Rev F.

11. Software Risk Management

Software risk management follows IEC 62304 clause 7 and IEC/TR 80002-1. The software system is Class B. The classification depends on hardware controls that do not rely on software: the normally-closed laser shutter interlock, the hardwired HV warning lamp and HV removal, mechanical relief valves, keyed fluid connectors and containment hardware. Where software is itself a risk control (for example the QC lockout), external controls (assay controls and result review) limit the harm, which keeps the item at Class B. SI-14 (fluid level monitoring) is Class A: its failure cannot contribute to a hazardous situation because the capacity check (SI-04) and the pressure check are independent of it.

11.1 Software causes of hazardous situations

Software item	Failure (assumed to occur)	Hazardous situation	Control outside the failing item / correction	Requirement
SI-11	Late sort decision under load	HS-07.1	Independent timing check; FIFO separation (ANM-0003)	FW-204
SI-13	Clog not detected	HS-04.2	Hardware drawer closes on breakoff loss; operator alarm	FW-210
SI-10	Fault manager fails to enter safe state	HS-02.3	SI-15 independent HV cut	FW-604
SI-15	Nuisance trip at boot	— (availability)	Boot grace with HV held off (ANM-0007)	FW-103
SI-05	QC window computed wrongly	HS-08.3	UTC computation; unit tests (ANM-0025)	RA-803
SI-20	Configuration rule bypassed	HS-08.1	Single write path; rules on all paths (ANM-0026)	AD-105
SI-21	Calibration applied to wrong laser	HS-08.2	Keyed by serial (ANM-0016)	AD-202
SI-41	Stale state after reconnect	HS-02.3	Full state read on reconnect (ANM-0023)	FW-805
SI-32	Audit trail incomplete	HS-08.4	Snapshot before edit (ANM-0028)	DM-201
SI-10	Fan control ignores laser temperature	HS-08.2, HS-17.2	FW-311 (CAPA-2028-014)	FW-311
SI-02	Lost events not reported	HS-08.6	Lost-event counter and warning	RA-304
SI-04	Capacity check wrong	HS-15.1	Unit tests; level sensor calibration	RA-207

The SI-10 fan-control finding is new in Rev D: the Rev C software failure analysis considered fan failure but not fan control that ignores a component temperature. The analysis checklist (RM-CHK-003 Rev C) now asks for every thermal control loop which temperature it regulates and whether supplier parts can change their thermal behavior.

Software requirements derived from hazards: 41 links from 8 hazards to SRS-SW-CSI-010. SOUP anomalies are evaluated in the SOUP register; in 2028 two SOUP vulnerabilities were patched (Section 14).

12. FMEA Summaries

The FMEAs rate severity (S, 1 to 5) using the harms catalogue and occurrence (O, 1 to 5) on the same scale as Section 4.2. Only failure modes with $S \geq 3$ or that feed a hazardous situation in the review region are shown; the full worksheets are DFMEA-CSI-001 Rev D (212 lines) and PFMEA-CSI-001 Rev B (148 lines). Failure modes are linked to the hazardous situation they can cause, so that FMEA occurrence estimates feed P1 in Section 6.

12.1 Design FMEA

Item	Failure mode	Effect	Cause	S	O before	Controls	O after	HS
SS-04 Laser module	Power drop at high base temperature	Sensitivity loss	Supplier TEC firmware change; fan control	4	3	FW-311 fan control; incoming hot test; thermal margin SYS-327	1	HS-08.2
SS-04 Launch plate	Spot shift after transport	Sensitivity loss	Crate resonance	4	3	Validated crate (ECO-0197); installation QC	1	HS-08.2
SS-03 Flow cell	Gel void	High CVs	Dispense volume	3	3	Validated process (ECO-0187); FAT CV test	1	HS-08.2
SS-05 Detector array	Wrong filter in position	Channel misassignment	Kitting error	4	3	Kitting scan; FAT spectral check; rules	1	HS-08.1
SS-06 PMT HV board	HV drift	Gain drift	Divider tolerance	3	3	Readback calibration (ANM-0034)	1	HS-08.2
SS-07 Deflection HV	Insulation breakdown	Shock	Saline ingress	4	2	Sealed feedthroughs; hipot; readback	1	HS-02.2
SS-07 Charge electrode	Timing error	Low purity	FPGA back-pressure	3	3	Separated FIFO (ANM-0003)	1	HS-07.1
SS-09 Chamber seal	Leak	Aerosol escape	Seal seating	4	3	Seal design; containment test each unit	1	HS-04.1
SS-01 Regulator	Fails open	Overpressure	Diaphragm failure	3	2	Relief valve	1	HS-09.1
SS-02 Loading port	Motor stall	Pinch / stuck tube	Mechanical wear	2	3	Stall detection	2	HS-06.1
SS-10 Workstation	Crash during sort	Loss of record	OS fault	3	3	Firmware autonomy; transactional DB	2	HS-15.2
SS-01 Level sensor	Reads full when empty	Run-out	Contamination	3	3	Calibration; redundant pressure check	1	HS-15.1

12.2 Process FMEA

Process step	Failure mode	S	O before	Controls (prevention; detection)	O after	HS
WI-CSI-1002 Gel coupling	Void in gel	3	3	Validated dispense and cure; void inspection	1	HS-08.2
WI-CSI-1003 Launch plate	Spot out of tolerance	4	3	Validated alignment; FTS §3	1	HS-08.2
WI-CSI-1005 Detector arrays	Wrong filter	4	3	Kitting scan; FAT spectral check	1	HS-08.1
WI-CSI-1007 HV assembly	Insulation damage	4	2	Validated assembly; 100% hipot	1	HS-02.2
Chamber sealing	Seal mis-seated	4	3	Containment test each unit	1	HS-04.1
WI-CSI-1009 Software install	Wrong version	4	2	Checksum readback	1	HS-08.1
Incoming inspection, lasers	Unapproved firmware accepted	4	3	Firmware check; hot test (CAPA CA3)	1	HS-08.2
WI-CSI-1010 FAT	Thermal drift not detected	4	3	6 h thermal soak (CAPA CA2)	1	HS-08.2
Labeling	Missing laser label	4	2	Label map per build (ECO-0193)	1	HS-01.2
WI-CSI-1012 Packing	Inadequate crating	3	3	Validated crate	1	HS-08.2

Incoming inspection of lasers and the 6 h thermal soak were added by CAPA-2028-014 (CA3, CA2). They address a gap in the Rev C PFMEA, which assumed that supplier parts are unchanged between approved revisions.

13. Use-Related Risk Analysis

Use-related risk is managed under IEC 62366-1 and recorded in HFE-CSI-001. Each critical task was analyzed for perception (P), cognition (C) and action (A) errors. The summative study VAL-CSI-03 used 15 operators and 15 supervisors (30 participants), production-equivalent instruments and final labeling. Results given as "after ANM-xxxx" refer to a repeat study after a design change.

Task	Critical task	Potential use error	PCA	HS	Controls	Summative result
T-01	Load sample tube	Hand in port while rising	A	HS-06.1	Force limit; e-stop; label	0 of 30
T-02	Recover from clog	Open door before evacuation	C	HS-04.2	Guided procedure; lockout of door step	0 of 30 (after ANM-0051)
T-03	Change collection tubes	Open collection door during sort	A	HS-05.1	Mechanical interlock	0 of 30
T-04	Clean aspirator	Reach in with HV on	P	HS-02.1	Warning light; IFU step	0 of 30
T-05	Select nozzle setup	Wrong size selected	P	HS-07.2	Nozzle marking; picture prompt	0 of 30 (after ANM-0060)
T-06	Set up drop delay	Manual delay wrong	C	HS-07.2	Automatic drop delay	0 of 30
T-07	Review daily QC	Proceed after failure	C	HS-08.3	Lockout	0 of 30
T-08	Run IVD assay	Edit template	C	HS-08.4	Locked template	0 of 30
T-09	Refill fluids	Connect wrong container	A	HS-14.1	Keyed connectors	0 of 30
T-10	Decontaminate	Skip bleach cycle	C	HS-04.4	Guided cleaning; certificate	0 of 30
T-11	Enter specimen ID	Wrong ID	A	HS-17.1	Review and release; LIS	1 of 30 (caught at review)
T-12	Sort rare population	Rate too high	C	HS-07.3	Advisory; IFU	0 of 30 (supplementary)

T-11 is the only task with an observed use error: one participant entered a transposed specimen ID, which was caught at the release step by the supervisor. The residual risk is carried in HS-17.1. Barcode entry of specimen IDs is planned for software release 3.1 (DCR-2029-004) and will reduce P1 further.

14. Cybersecurity Risk

Security risk is assessed with a STRIDE threat model (IEC 81001-5-1, AAMI TIR57). Exploitability is scored with CVSS 3.1 before and after controls. Threats that can affect safety or effectiveness are carried into HA-16 and HS-17.2 in Section 6. Controls are verified by DVP-CSI-701, including an independent penetration test.

Threat	Description	Attack vector	CVSS before	HS	Controls	CVSS after
TH-01	Replay of captured command	Adjacent network	7.5	HS-16.1	Session keys, sequence numbers	2.1
TH-02	Malicious firmware update	Local / supply chain	8.2	HS-16.1	Signed packages, manifest	2.4
TH-03	Credential guessing	Local	6.5	HS-08.4	Lockout, complexity, audit	2.6
TH-04	Ransomware on workstation	Network	8.8	HS-17.2, HS-16.2	Locked image, isolation, backup	3.1
TH-05	Theft of workstation	Physical	5.9	HS-16.2	Encryption at rest	1.8
TH-06	Tampering with exported files	Network	6.1	HS-16.1	Checksums	2.0
TH-07	Vulnerable SOUP component	Network	7.8	HS-16.1	SBOM monitoring, patching	3.0

Post-market: SBOM monitoring flagged 2 SOUP vulnerabilities in 2028 (web rendering library, CVSS 6.1; compression library, CVSS 5.3). Neither was reachable from the isolated instrument link; both were patched in the 3.0.3 maintenance release within the 30-day target.

15. Risks from Anomalies Found in V&V

Every anomaly in ADL-CSI-001 linked to a hazard was reviewed for its effect on the risk estimate and on the risk controls. Anomalies are linked by hazard; the affected hazardous situations are listed.

Anomaly	Hazard (HS)	Effect on risk	Outcome
ANM-0003 Sort decision latency exceeds 20 μ s above 95,000 events/s	HA-07 (HS-07.1, HS-07.2, HS-07.3)	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0007 Safety supervisor removes HV enable during normal boot	HA-02 (HS-02.1, HS-02.2, HS-02.3)	No change to risk estimate	Corrected and re-verified; risk estimate unchanged after correction
ANM-0011 Clog response exceeds 2 s in 2 of 59 trials	HA-04 (HS-04.1, HS-04.2, HS-04.3, HS-04.4)	Risk control did not meet requirement	Corrected and re-verified; risk estimate unchanged after correction; HA-CSI-001 updated (control added)
ANM-0012 PI CV 3.1% on DV-02 on day 2	HA-08 (HS-08.1, HS-08.2, HS-08.3, HS-08.4, HS-08.5, HS-08.6)	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0016 Laser calibration factor applied to wrong laser on shared beam spot	HA-08 (HS-08.1, HS-08.2, HS-08.3, HS-08.4, HS-08.5, HS-08.6)	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction

Anomaly	Hazard (HS)	Effect on risk	Outcome
ANM-0021 Radiated immunity: event rate spikes at 3 V/m, 80 to 120 MHz	HA-13 (HS-13.1)	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0023 After link loss, Research App shows Ready while firmware is in safe state	HA-02 (HS-02.1, HS-02.2, HS-02.3)	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0025 QC lockout window 25 h across daylight saving change	HA-08 (HS-08.1, HS-08.2, HS-08.3, HS-08.4, HS-08.5, HS-08.6)	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0026 480/20 filter rule not enforced after field upgrade to 445 nm	HA-08 (HS-08.1, HS-08.2, HS-08.3, HS-08.4, HS-08.5, HS-08.6)	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0030 Replayed command accepted after reconnect (penetration test)	HA-08 (HS-08.1, HS-08.2, HS-08.3, HS-08.4, HS-08.5, HS-08.6)	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0032 Laser warning label missing on rear panel of 5-laser build	HA-01 (HS-01.1, HS-01.2, HS-01.3)	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0040 DV-05 needed realignment after ISTA 3E transport test	HA-08 (HS-08.1, HS-08.2, HS-08.3, HS-08.4, HS-08.5, HS-08.6)	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0048 Duplicate of ANM-0023	HA-02 (HS-02.1, HS-02.2, HS-02.3)	Requirement not met	Duplicate; evaluated under the original anomaly
ANM-0051 Usability verification: clog recovery step skipped	HA-04 (HS-04.1, HS-04.2, HS-04.3, HS-04.4)	No change to risk estimate	Corrected and re-verified; risk estimate unchanged after correction; HA-CSI-001 updated (control added)
ANM-0056 VAL-CSI-02 Site 3: CD34+ purity 93% on two specimens	HA-07 (HS-07.1, HS-07.2, HS-07.3)	Requirement not met	Corrected and re-verified; risk estimate unchanged after correction
ANM-0060 VAL-CSI-03: users unsure which nozzle is installed	HA-08 (HS-08.1, HS-08.2, HS-08.3, HS-08.4, HS-08.5, HS-08.6)	No change to risk estimate	Corrected and re-verified; risk estimate unchanged after correction; HA-CSI-001 updated (control added)
ANM-0066 Plate voltage readback drift warning after 6 h sorting	HA-02 (HS-02.1, HS-02.2, HS-02.3)	No change to risk estimate	Corrected in board FPGA 3.0.6 (maintenance release, Sept 2027); closed; risk estimate unchanged

ANM-0062 (violet detector QC failures, found at a beta site) was classified as a performance anomaly without a safety link at the time. The same failure mode recurred in the field in 2028 and led to CAPA-2028-014. The anomaly triage procedure now requires a risk review for every QC failure that is not explained by a user action.

16. Aggregated Risk per Harm

Individual residual risks that lead to the same harm are added to check that the combined probability of each harm is still acceptable (ISO/TR 24971 clause 8). The sum is conservative because the situations are not mutually exclusive.

Harm	S	Hazardous situations	Sum of residual P	Level	Evaluation
H-01 Permanent retinal injury	S4	HS-01.1, HS-01.2	5.1×10^{-5}	P1	Acceptable
H-02 Minor skin or eye irritation from stray light	S2	HS-01.3	10^{-5}	P1	Acceptable
H-03 Electric shock injury from high voltage	S4	HS-02.1, HS-02.2, HS-02.3	1.6×10^{-4}	P2	Review (benefit-risk)
H-04 Electrocution (death)	S5	HS-12.1	10^{-7}	P1	Review (benefit-risk)
H-05 Laboratory-acquired infection	S4	HS-04.1, HS-04.2, HS-04.3, HS-04.4, HS-05.1	8.6×10^{-5}	P1	Acceptable
H-06 Pinch or minor laceration	S2	HS-06.1, HS-06.2	6×10^{-5}	P1	Acceptable
H-07 Crush injury	S3	HS-11.1	3×10^{-5}	P1	Acceptable
H-08 Chemical irritation of skin or eyes	S2	HS-10.1	10^{-5}	P1	Acceptable
H-09 Impact or spray injury from pressure release	S3	HS-09.1, HS-09.2	5×10^{-5}	P1	Acceptable
H-10 Misdiagnosis of a hematological disease	S4	HS-07.1, HS-07.2, HS-07.3, HS-08.1, HS-08.2, HS-08.3, HS-08.4, HS-08.5, HS-13.1, HS-16.1, HS-17.1	7.9×10^{-4}	P2	Review (benefit-risk)
H-11 Delayed result (hours to days)	S2	HS-17.2	5×10^{-4}	P2	Acceptable
H-12 Invasive specimen re-collection	S3	HS-14.1, HS-15.1, HS-15.2	1.2×10^{-3}	P3	Review (benefit-risk)
H-13 Incorrect cell therapy product characterization	S4	HS-08.6	10^{-4}	P2	Review (benefit-risk)
H-15 Disclosure of patient information	S2	HS-16.2	10^{-5}	P1	Acceptable
H-16 Burns or smoke inhalation from fire	S4	HS-12.2	10^{-7}	P1	Acceptable

No harm is unacceptable in aggregate. 5 harms are in the review region in aggregate (H-03, H-04, H-10, H-12, H-13). For H-10 (misdiagnosis), the largest contributor is rare-event sorting at high event rates (HS-07.3), followed by HS-08.1, HS-08.2 and HS-17.1; all are addressed in Section 18.

17. Overall Residual Risk Evaluation

The overall residual risk was evaluated on 2028-12-08 by a panel of Quality, Systems Engineering, Clinical Affairs, Human Factors and the independent clinical flow cytometry reviewer, using the method in RMP-CSI-001 Section 9:

- Review of each individual residual risk (Table 6-2) and the verification of its controls (Section 7).
- Aggregated risk per harm (Section 16).
- Combined exposure of one person in one task: an operator clearing a clog is exposed at the same time to aerosol (HS-04.2), high voltage (HS-02.1) and moving parts (HS-06.1). The clog response removes HV and closes the drawer before the recovery prompt, so the exposures do not coincide.
- Comparison with legally marketed droplet cell sorters: labeling, published biosafety incidents and recalls in the FDA MAUDE and recall databases 2019 to 2028 (search record LIT-CSI-2028-01).
- Post-market experience of the CSI-100 in 2028 (Section 21), which confirmed the estimates for 5 of the 7 monitored situations and led to the changes for HS-08.2 and HS-17.2.
- Residual risks disclosed in labeling (Section 19).

Conclusion: no individual or aggregated residual risk is unacceptable. 8 individual residual risks are in the review region and are justified in Section 18. The overall residual risk is acceptable in relation to the benefits of the intended use and is comparable to or lower than that of similar devices on the market.

18. Benefit-Risk Analysis

18.1 Medical benefits

- Identification and enumeration of cell populations for the diagnosis and monitoring of leukemias, lymphomas and immune disorders, with performance shown in method comparison and multi-site precision (VAL-CSI-01, VAL-CSI-04).
- Physical isolation of rare populations at high purity (above 98% in VAL-CSI-02) for molecular testing of the sorted cells, which increases the sensitivity of downstream tests compared with testing of unsorted specimens.
- Single-cell deposition with index data for clonality studies in the clinical laboratory.
- Locked assay templates, automated setup and QC lockout reduce operator-dependent variation compared with manual gating and manual drop-delay setup on earlier sorters.

18.2 Residual risks in the review region

HS	Residual risk	Why further reduction is not practicable; justification
HS-02.1 S4 P2	HV contact when cleaning the aspirator with plates energized.	Plates stay energized with the door open because the operator must see the deflected streams during setup. A door interlock that removed HV would prevent stream setup. Warning lamp driven from the HV supply, recessed plates, IFU step. Same approach as comparable sorters; 0 events in 2028.
HS-07.3 S4 P2	Purity below user expectation for populations below 1% at high event rates.	Coincidence is a physical limit of droplet sorting at a given event rate. Lowering the maximum rate for all users would remove benefit for common sorts. Advisory in software and event rate guidance in the IFU.
HS-08.1 S4 P2	Wrong optical configuration after a field upgrade.	Configuration rules and board inventory check block seeded errors in 100% of tests; remaining probability is from errors outside the rule set. Assay controls provide a further, independent check.
HS-08.2 S4 P2	Sensitivity drift between QC and a later run.	Drift cannot be excluded for every component over the product life. CAPA-2028-014 added fan control, a temperature warning and thermal margin. Assay controls with every run remain the final check. Estimate to be re-checked at 12 months (Section 8.2).
HS-08.6 S4 P2	Incorrect CD34+ enumeration.	Rare-event counting has an inherent statistical limit. Locked template with minimum event count; lost events reported; release testing includes independent checks.
HS-12.1 S5 P1	Death from electric shock with an enclosure fault.	Design meets IEC 61010-1 with two independent means of protection and 100% production testing. Probability improbable; S5 severity places any mains-powered instrument in this cell.
HS-15.2 S3 P3	Loss of an irreplaceable specimen when a sort stops.	Clogs cannot be fully prevented with biological specimens. Sample tube is unloaded on clog, which preserves the remaining specimen, and the sort can be resumed.
HS-17.1 S4 P2	Result reported for the wrong patient after a specimen ID entry error.	Specimen identification is controlled by laboratory procedures. The instrument supports release with ID confirmation and LIS export without transcription. Barcode entry is planned for release 3.1.

18.3 Conclusion

The medical benefits of the CSI-100 when used as intended outweigh the overall residual risk. Each residual risk in the review region has been reduced as far as possible without adversely affecting the benefit-risk ratio and is disclosed to users.

19. Disclosure of Residual Risks

Residual risks are disclosed in the labeling (ISO 14971 clause 8). Information for safety that is itself a risk control was verified for effectiveness in VAL-CSI-03.

HS	Residual risk disclosed	Where disclosed
HS-01.1	Direct or specularly reflected beam reaches the operator's eye.	OG-CSI-001 §2.3
HS-01.2	Service engineer exposed to open Class 3B beams.	SM-CSI-001 §3
HS-02.1	Hand contacts an energized plate.	OG-CSI-001 §6.4
HS-04.1	Operator in the room inhales aerosol.	OG-CSI-001 §3
HS-04.2	Operator's face near a chamber full of aerosol.	OG-CSI-001 §7.2
HS-04.3	Specimen splashes onto operator's gloves or face.	OG-CSI-001 §5.1
HS-04.4	Contact with residual specimen material.	SM-CSI-001 §2
HS-05.1	Hand near energized plates and aerosol.	OG-CSI-001 §7.1
HS-06.1	Finger trapped between port and chamber.	Label
HS-07.1	Sorted fraction reported pure is used for molecular testing (e.g., mutation analysis).	OG-CSI-001 §7.4
HS-07.3	Coincident events lower purity below user expectation.	OG-CSI-001 §7.4
HS-08.2	Dim marker population under-detected in afternoon runs.	OG-CSI-001 §9 (Rev D)
HS-08.3	Unverified instrument generates clinical result.	OG-CSI-001 §9
HS-08.6	Graft dose miscalculated.	Assay IFU
HS-09.2	Lid released under pressure.	OG-CSI-001 §5.3
HS-11.1	Instrument tips.	SPG-CSI-001
HS-16.1	Results or settings altered undetected.	Security documentation
HS-16.2	Patient data disclosed.	Security documentation
HS-17.1	Wrong result in patient record.	OG-CSI-001 §10
HS-17.2	Result delayed by one or more days.	OG-CSI-001 §9

OG-CSI-001 Rev D (December 2028) adds to §9 the laser temperature warning, its meaning and the action to take, and a statement that assay controls must be run with every IVD run even when QC has passed.

20. Risk Management Report

This section is the risk management report RMR-CSI-001 Rev B (ISO 14971 clause 9), the first annual review after commercial release. Rev A was approved for release in May 2027.

Review question	Finding	Evidence
Has the risk management plan been implemented appropriately?	Yes	Sections 2 to 16; RMP-CSI-001 Rev C
Is the overall residual risk acceptable?	Yes	Sections 16, 17 and 18
Are methods in place to collect and review production and post-production information?	Yes. They worked in 2028: log trending detected the thermal drift before any wrong result was reported	Section 21; PMS-CSI-001 Rev B
Have all risk controls, including those added in Rev D, been implemented and verified?	Yes	Section 7; CAPA-2028-014 E1 to E3

Review question	Finding	Evidence
Have risks arising from controls been evaluated?	Yes; one new situation (HS-17.2)	Section 10.1
Did post-market data change any estimate?	Yes: HS-08.2 P1 revised; HS-17.2 added	Sections 8.2, 21.2
Open items	Re-check HS-08.2 at 12 months (PM-2029-02); barcode entry (DCR-2029-004)	Section 21.4

Conclusion: the risk management process for the CSI-100 meets ISO 14971:2019; the overall residual risk remains acceptable; continued distribution is supported.

Reviewer	Role	Signature / Date
Quality Engineering	Risk management owner	
Clinical Affairs	Clinical assessment	
Systems Engineering	Technical lead	
Independent reviewer	Clinical flow cytometry expert	
Top management	Approval	

21. Production and Post-Production Activities

21.1 Information collected

Source	Content	Review frequency	2028 volume
Complaints and adverse events	Customer complaints; MDR and vigilance reports	Each event; monthly trend	73 complaints; 0 reportable
Service data	Repairs, replaced parts, field alignment results	Monthly	196 service visits
Instrument internal logs	Safe-state entries, clog events, HV faults, QC results, laser temperatures (FW-701 to FW-704, FW-312)	Quarterly	64 instruments
Production data	NCRs, first-pass yield, FAT failures	Monthly	38 units built; FPY 92%
Cybersecurity monitoring	SBOM vulnerabilities; security incidents	Monthly	2 SOUP vulnerabilities; 0 incidents
External sources	Literature, adverse event and recall databases, standards	Annually	LIT-CSI-2028-01

21.2 Estimated versus observed (2028)

HS	Situation monitored	Estimated P (residual)	Observed in 2028	Conclusion
HS-04.2	Clog followed by door opening before evacuation	2×10^{-5}	0 events in 212 clog recoveries (logs)	Estimate confirmed
HS-08.2	Sensitivity drift after passing QC	1×10^{-3} (Rev C residual)	7 of 64 instruments with thermal drift (2028)	P1 revised to 10^{-1} before CAPA; CAPA-2028-014; residual 10^{-3}
HS-17.2	Result delayed by QC failures	Not analyzed (Rev C)	41 QC-blocked events at 7 sites	New situation added in Rev D
HS-02.1	HV contact during cleaning	1.5×10^{-4}	0 complaints; 0 HV fault logs with door open	Estimate confirmed
HS-06.1	Pinch at loading port	5×10^{-5}	1 complaint (no injury)	Estimate confirmed
HS-15.1	Sheath run-out during sort	2×10^{-4}	0 events; 318 blocked sort starts	Control effective
HS-16.1	Unauthorized modification	10^{-6}	0 incidents; 2 SOUP vulnerabilities patched	Estimate confirmed

Observed rates are compared with the estimate using the 95% upper bound where no events occurred. A situation is escalated when the observed rate exceeds the estimate by one probability level or when a harm occurs.

21.3 CAPA-2028-014 impact on this file

CAPA-2028-014 was opened in August 2028 after 7 instruments with 405 nm lasers from a new supplier lot failed violet QC in the afternoon. The investigation found a supplier TEC firmware change and fan control that followed room temperature only. The risk file was changed as follows: HS-08.2 P1 revised (Section 8.2); HS-17.2 added for loss of availability (Section 10.1); FW-311, RA-107 and SYS-327 added as controls; DFMEA and PFMEA lines added for supplier firmware changes and thermal soak; the software failure analysis checklist extended (Section 11). The CAPA effectiveness checks E1 to E3 were met on 2028-11-30.

21.4 Open post-market actions

Action	Description	Due
PM-2029-02	Re-check HS-08.2 residual P1 with at least 60 instrument-years of data after the fan control update	Dec 2029
PM-2029-03	Monitor QC-blocked days (HS-17.2); escalate if more than 1 per instrument-year	Quarterly
DCR-2029-004	Barcode entry of specimen IDs (HS-17.1)	Release 3.1
PM-2029-05	Supplier change notification clause added to quality agreements for all laser and detector suppliers	Jun 2029

Triggers for updating this file: a new hazard or hazardous situation; a harm occurring more often or with higher severity than estimated; a failed or bypassed risk control; a design, supplier or process change; a change in the state of the art or in applicable standards.

22. Reference Documents

Document	Revision	Title
ISO 14971	2019	Application of risk management to medical devices
ISO/TR 24971	2020	Guidance on the application of ISO 14971
IEC 62304; IEC/TR 80002-1	2015 (A1); 2009	Software life cycle; guidance on applying ISO 14971 to software
IEC 62366-1	2020 (A1)	Usability engineering
IEC 81001-5-1; AAMI TIR57	2021; 2016 (R2023)	Security in the product life cycle; security risk management
IEC 61010-1, -2-101; IEC 60825-1; IEC 61326-2-6	Current editions	Electrical, laser and EMC safety
QP-RM-001	Rev F	Risk Management Procedure and Risk Acceptability Policy
RMP-CSI-001	Rev C	Risk Management Plan, CSI-100
HA-CSI-001	Rev D	Hazard Analysis worksheets
FTA-CSI-001	Rev B	Fault Tree Analysis
DFMEA-CSI-001; PFMEA-CSI-001	Rev D; Rev B	Design and Process FMEA
TM-SEC-CSI-001	Rev C	Threat Model
HFE-CSI-001	Rev B	Human Factors Engineering Report
SRS-CSI-001; SRS-SW-CSI-010	Rev E; Rev F	System and Software Requirements Specifications
SAD-SW-CSI-011	Rev C	Software Architecture Description
VVP-CSI-001; TM-CSI-DT-01	Rev A; Rev C	V&V Plan; Requirements Traceability Matrix
ADL-CSI-001	Export 07	Anomaly and Defect Tracking Log
CAPA-2028-014	Closed	CAPA Report: violet detector QC failures
DTP-CSI-001	Rev B	Design Transfer Plan
PMS-CSI-001; PMS-PRED-001	Rev B; Rev A	Post-Market Surveillance Plan; predicate instrument data summary
OG-CSI-001; SM-CSI-001; LBL-CSI-001	Rev D; Rev C; Rev C	Operator's Guide; Service Manual; Labeling Specification
LIT-CSI-2028-01	Rev A	Literature and adverse event database search